

Identifying Bones and Shell

When a zooarchaeologist begins to analyze a bag of recovered faunal material from a stratigraphic unit of an excavation, the bag will contain a large variety of types of bones (and perhaps also shells) mixed together. To identify them, the zooarchaeologist will group similar bones together and then work to identify each. A common approach is to separate bones first by class: mammal, fish, bivalve, reptile, bird, etc. Even small pieces of bone can typically be distinguished at this level. Next, the zooarchaeologist will focus on one of these piles and look at each piece in turn to recover **primary data**. These are characteristics of the bone itself, including the following:

- What element (part of the body) is it? Which side?
- To what species does it belong? If species is uncertain, in what **taxonomic group** should it be placed?
- How many are present (count)?
- How large is it?
- Is it broken or entire? If broken, which part of the bone is it?
- How has the bone been modified? Is it cut (and by what) or burned?
- Can the bone be used to estimate age or sex? If so, what age or sex is indicated?

Primary data are data obtained from the physical characteristics of individual bones themselves, such as element, side, **taxon**, count, and modification.

Taxon (plural: taxa) is the taxonomic group of an individual organism, whether that is species or a higher classification. For example, humans (*Homo sapiens*) are part of the family Hominidae (hominids, including gorillas, chimps, bonobos, and orangutans), the order Primates (all primates), and the class Mammalia (mammals). Often, closely related taxa—

such as domesticated sheep (*Ovis aries*) and goats (*Capra hircus*)—have bones that are hard to distinguish, so they are placed into a higher taxonomic group during identification: Ovis/Capra, or sheep/goat.

Counting Bones

Bones are typically counted using the metric **NISP** (number of identified specimens), which is the basic count of all bones or bone fragments found. If there are 1,204 fragments of bone from a single stratigraphic unit, the sum of the NISP of all taxa from that unit (including unknown or indeterminate bones) should be 1,204. A second metric commonly used is **MNI** (minimum number of individuals). For each taxon, the MNI is the minimum number of individual animals that could have produced the assemblage. This number is based on the NISP by specific skeletal element for that taxon. Test yourself using the example below.

NISP stands for number of identified specimens and is the count of all bones or bone fragments by taxon for a sample, stratigraphic unit or site.

MNI stands for minimum number of individuals and is the minimum number of individual animals that could have contributed to produce the zooarchaeological assemblage under investigation.

Test Yourself

You just completed the recording of primary data on a zooarchaeological assemblage from St. Augustine, Florida, c. 1700 CE. You have recorded the NISP for all white-tailed deer (*Odocoileus virginianus*) bones, which are tabulated below by element and side for this taxon. (Note: If you are unsure what part of the body a bone is from, just Google its name!)

Element	NISP Left	NISP Right	NISP Total
Mandible	2	2	4
Vertebrae			2
Scapula	3	2	5
Humerus	3	2	5
Radius	1	5	6
Ischium	0	3	3
Femur	4	0	4
Tibia	1	3	4
Astragalus	2	2	4

What is the minimum number of individuals of white-tailed deer represented at this site?

☒ A. **Correct:** 5 Correct answer

☐ B. 6

☐ C. 21

☐ D. 37

Correct! The element with the greatest number of unique bones is the radius (lower arm bone) with 5 right radii. This is the minimum number of deer that could have produced this assemblage.

Ageing and Sexing Bones

To determine the age at death and sex of an individual from its skeletal anatomy alone can be difficult. Instead, we are only able to estimate these from a single bone or group of bones. This is due to differences in development among individuals. For example, in many species of animals, males are much larger than females, a phenomenon termed “sexual dimorphism.” But while measuring specific bones can distinguish a population of larger bones from one of smaller bones, which would suggest a difference in sex, often there are bones of intermediate size, and it is impossible to know from skeletal anatomy alone whether one of the larger bones is definitively male. Better characteristics come from skeletal elements important in childbirth—the male pelvis is often differently shaped from the female pelvis for this reason—or from secondary sexual characteristics, such as horns, that distinguish the sexes. Still, even these are necessarily estimates; genotyping the individual to determine its chromosomal complement is the one way to be certain of its sex.

Age at death can be determined through bone fusion or the eruption and wear of teeth. Humans and other mammals have developmentally programed schedules for the growth, eruption, and loss of both juvenile and adult teeth. If we found a human skeleton with only two adult teeth—the upper incisors—and the remaining teeth still juvenile, we might place the age at death around age eight, but as we know, the age of appearance of those teeth can vary among individuals by a few years. We can do the same with animals. Since most have shorter lifespans than humans, the range of ages at which specific dental patterns appear is more constrained. See how this works for white-tailed deer at [this website](https://extension.missouri.edu/publications/g9484) (<https://extension.missouri.edu/publications/g9484>) of the University of Missouri Extension. Note that age uncertainty is narrower at young ages (e.g., 2.5–4 months, 18–19 months) than at adult ages (over 2.5 years), at which dental age estimates are based on the wear on teeth, since at this point, all adult teeth are fully erupted.